PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – IV
DEFLECTIONS

Syllabus : Control of deflections – factors influencing deflections – short-term deflections of un-cracked beams – prediction of long-time deflections.

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Exam tip : Learn the deflection formula for each tendon profile ($Pe\,L^2/8EI$, $5PeL^2/48EI$, $PeL^2/12EI$) – every numerical uses them.

1. Introduction

PSC members are slender and made of high-strength materials – so **serviceability (deflection)** often governs the design rather than strength.

- Prestress produces an **upward deflection (camber)** that opposes the downward deflection due to loads; the net deflection may be upward or downward.
- Excess camber or deflection leads to: cracking of partitions / finishes, ponding on roofs, drainage problems, vibration, machine misalignment, poor appearance and loss of user confidence.
- Deflection is a **serviceability limit state** – checked under working loads (and for long-term effects).

IS 1343 / IS 456 limits (cl. 23.2)

Case	Limit
Final deflection (due to all loads, incl. creep, shrinkage, temperature)	≤ span / 250
Deflection after erection of partitions & finishes (incl. creep, shrinkage, temperature)	≤ span / 350 or 20 mm, whichever is less
Cantilevers	span replaced by 2 × projection
Camber (upward) – practical	excess camber to be avoided (≈ span/300 – span/250 at transfer)

Camber / deflection in a PSC beam

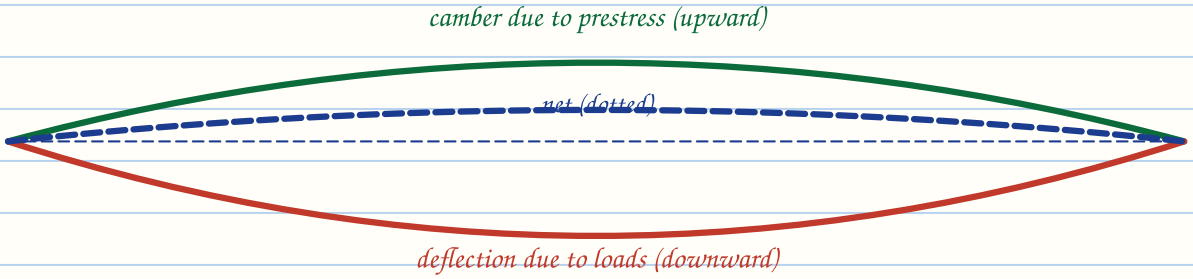


Fig. Upward camber from prestress opposes downward load deflection

Net deflection = (deflection due to loads) – (camber due to prestress). At transfer only self-weight acts while P is maximum → **maximum camber** occurs at transfer.

2. Factors Influencing Deflections

Factor	Effect on deflection
Magnitude of prestressing force P	$\uparrow P \rightarrow \uparrow$ camber (upward deflection)
Eccentricity & profile of tendon	Large e at mid-span, parabolic/bent profile → more camber; shape decides the coefficient (1/8, 5/48, 1/12)
Load : self weight, dead, live; their type and duration	Deflection ∝ load; sustained loads increase with creep
Span L	Short-term $\delta \propto L^2$ (prestress) or L^4 (UDL) – increases rapidly with span
Flexural rigidity $E_c I$	Higher grade concrete → $\uparrow E_c$; deeper section → $\uparrow I$; cracked section → lower I
Modulus of elasticity of concrete E_c (age)	$E_c = 5700\sqrt{f_{ck}}$ (IS 1343) ; E_c at transfer is smaller (f_{ci}) → larger camber at transfer
Creep of concrete	Deflection increases with time (factor $1+\theta$) ; camber also grows with time
Shrinkage of concrete	Curvature (unsymmetric steel) → additional deflection
Loss of prestress (creep, shrinkage, relaxation)	Reduces camber with time → net deflection increases
Cracking of concrete	Reduces effective I → deflection suddenly increases after M_{cr}
Support conditions & continuity	Fixed ends / continuity reduce deflection ($M_{support}$ hogging)
Temperature & humidity, age at loading	Affect creep / shrinkage → long-term deflection
Type of prestressing / bond	Un-bonded vs bonded – changes stress in steel at overload

Summary : δ depends on **(P, e, profile) × (L^2/EI)** and the **load history** through creep. Controlling P, e and $E_c I$ controls δ.

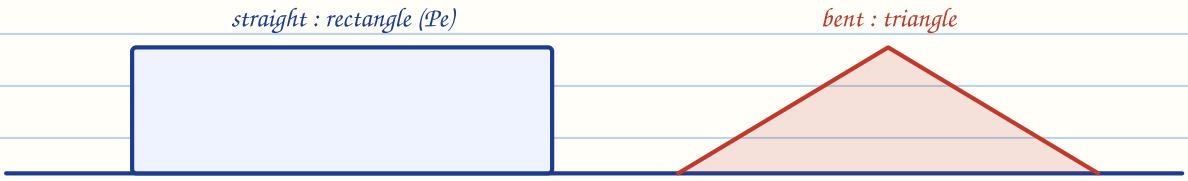
3. Deflection of Beams – Basic Formulae

All short-term deflection of un-cracked PSC beams is calculated by elastic theory with the gross (or transformed) $E_c I$: $d^2y/dx^2 = M/EI$.

Beam & load	Max. deflection
Simply supported, UDL w	$\delta = 5\,w\,L^4 / (384\,E\,I)$
Simply supported, central point load W	$\delta = W\,L^3 / (48\,E\,I)$
Simply supported, constant moment M	$\delta = M\,L^2 / (8\,E\,I)$
Cantilever, UDL w	$\delta = w\,L^4 / (8\,E\,I)$
Cantilever, end load W	$\delta = W\,L^3 / (3\,E\,I)$
Cantilever, end moment M	$\delta = M\,L^2 / (2\,E\,I)$

Moment-area method (for any tendon profile)

- Deflection at mid-span = moment of the M/EI diagram between the support and mid-span, taken about the support (for symmetrical loading) / tangential deviation.
- Prestress moment at section = $P \cdot e_x$ (e_x = eccentricity at x). The curvature diagram $M/EI = P\,e_x / EI$.
- Prestress M-diagram for parabolic tendon = parabola; for bent tendon = triangle; straight = rectangle.



M-diagrams of prestress (shape ~ tendon profile)

Fig. Bending-moment diagrams from different tendon profiles

Sign convention : **upward deflection (camber) +ve** for prestress effects, downward for load effects – subtract to get net.

4. Deflection due to Prestress (Short-term, Un-cracked)

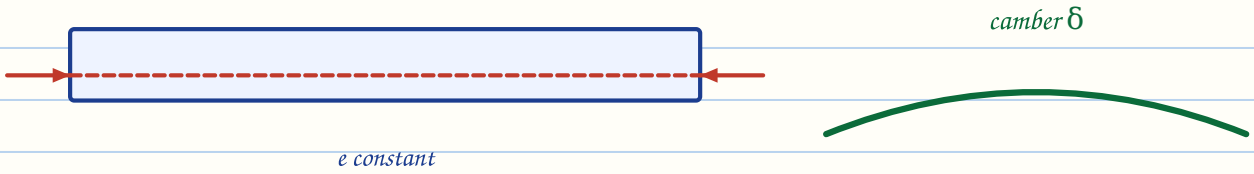
(a) Concentric straight tendon

$e = 0 \rightarrow$ no moment $\rightarrow$ **no deflection** (only shortening).

(b) Eccentric straight tendon (constant e)

Constant moment $M = P\,e$. Curvature $= Pe/EI$ (constant) $\rightarrow$ circular arc.

$$\delta = P\,e\,L^2/(8\,E\,I) \text{ (upward at mid-span)}$$



(c) Bent (harped) tendon – single kink at mid-span, e at mid, 0 at ends

- Equivalent upward load at mid-span $W = 4\,P\,e\,/\,L$; $\delta = W\,L^3/48EI$:

$$\delta = P\,e\,L^2/(12\,E\,I)$$

- For a **two-point bent tendon** (kinks at distance a from each support, e at the kinks) : each kink carries $W = P\,e/a$ upward and $\delta = P\,e\,(3L^2 - 4a^2)/(24\,E\,I)$ ($a = L/2$ gives back $PeL^2/12EI$ ✓).

(d) Parabolic tendon – e at mid-span, 0 at the ends

- Equivalent upward UDL $w_p = 8Pe/L^2$; $\delta = 5\,w_p\,L^4/384EI$:

$$\delta = 5\,P\,e\,L^2/(48\,E\,I)$$

(e) Parabolic tendon with eccentricity e_a at ends and e_c at mid-span

$$\delta = P\,L^2\,(e_a + 5\,e_c)/(48\,E\,I) = P\,e_a\,L^2/8EI + 5P(e_c - e_a)\,L^2/48EI$$

(e below C.G. +ve; e above C.G. –ve. For $e_a = 0 \rightarrow 5PeL^2/48EI$ ✓ ; for $e_a = e_c \rightarrow PeL^2/8EI$ ✓.)

Cantilever : straight tendon : $\delta = P\,e\,L^2/(2EI)\uparrow$; parabolic tendon (e at the fixed end, horizontal tangent at the free end) : $\delta = P\,e\,L^2/(12EI)\uparrow$. In every case integrate M/EI to get the coefficient.

5. Worked Examples (Short-term)

Example 1 – straight eccentric tendon

A PSC beam 150 × 300 mm, span 8 m, P = 300 kN at e = 50 mm (constant). E_c = 35000 N/mm². Self-weight + imposed load = 1.125 + 3 kN/m. Find camber, deflection and net deflection. Loss 15 %.

$$I = 150\times300^3/12 = 3.375\times10^8\ mm^4\ ;\ E_cI = 1.181\times10^{13}\ N\cdot mm^2$$

$$(i)\ Prestress: \delta_p = PeL^2/8EI = 300\times10^3\times50\times8000^2/(8\times35000\times3.375\times10^8) = 10.16\ mm\uparrow$$

$$(ii)\ Self\ weight: \delta_g = 5\times1.125\times8000^4/(384\times35000\times3.375\times10^8) = 5.08\ mm\downarrow$$

$$(iii)\ At\ transfer: net = 10.16 - 5.08 = 5.08\ mm\uparrow\ (camber)$$

$$(iv)\ Imposed\ 3\ kN/m: \delta_q = 13.54\ mm\downarrow;\ P_e = 0.85P \rightarrow \delta_p = 8.64\ mm\uparrow$$

$$(v)\ At\ service: net = 5.08 + 13.54 - 8.64 = 9.98\ mm\downarrow < L/250 = 32\ mm\ \checkmark$$

Example 2 – bent tendon

Beam 200 × 400, span 6 m, bent tendon with e = 80 mm at mid-span (0 at the ends), P = 500 kN, E_c = 36000.

$$I = 200\times400^3/12 = 1.0667\times10^9\ mm^4$$

$$\delta = PeL^2/(12EI) = 500\times10^3\times80\times6000^2/(12\times36000\times1.0667\times10^9) = 3.13\ mm\uparrow$$

Equivalent load $W = 4Pe/L = 4\times500\times0.08/6 = 26.7$ kN upward at mid-span ; check $\delta = WL^3/48EI = 26.7\times10^3\times6000^3/(48\times36000\times1.0667\times10^9) = 3.13$ mm ✓.

Example 3 – parabolic tendon, eccentric at the ends

Beam 300 × 600, span 12 m, M50 (E_c = 40305), P = 1000 kN. Tendon parabolic, 200 mm **above** C.G. at the ends and 200 mm **below** at mid-span. Find the camber.

$$I = 5.4\times10^9\ ;\ e_a = -200,\ e_c = +200$$

$$\delta = PL^2(e_a + 5e_c)/(48EI) = 10^6\times(12000)^2\times(-200 + 1000)/(48\times40305\times5.4\times10^9) = 11.0\ mm\uparrow$$

6. Example 4 – Complete deflection check (Design Ex. of Unit III)

Beam 300 × 500 mm, span 10 m, parabolic tendon (e = 120 mm at mid-span, 0 at supports), P_i = 1057 kN, losses 20 % (η = 0.8), self weight 3.75 kN/m, imposed 10 kN/m, M40 → $E_c = 5700\sqrt{40} = 36050$ N/mm². Find the short-term deflections at transfer and service.

$$I = 300\times500^3/12 = 3.125\times10^9 \text{ ; } E_cI = 1.1266\times10^{14} \text{ N·mm}^2$$

At transfer (P_i + self weight)

$$\delta_p = 5PeL^2/48EI = 5\times1057\times10^3\times120\times10^8/(48\times1.1266\times10^{14}) = 11.73 \text{ mm} \uparrow$$

$$\delta_g = 5wL^4/384EI = 5\times3.75\times10^{16}/(384\times1.1266\times10^{14}) = 4.33 \text{ mm} \downarrow$$

Net at transfer = 11.73 − 4.33 = **7.40 mm ↑** (camber ≈ L/1350 ✓ small)

At service (ηP_i + all loads, 13.75 kN/m)

$$\delta_p = 0.8\times11.73 = 9.38 \text{ mm} \uparrow \text{ ; } \delta_w = 4.33\times13.75/3.75 = 15.88 \text{ mm} \downarrow$$

Net = 15.88 − 9.38 = **6.50 mm ↓** < L/250 = 40 mm ✓

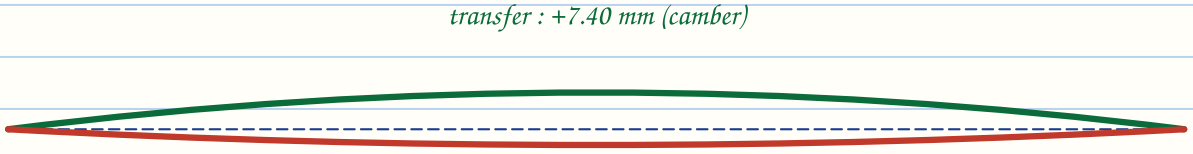


Fig. Net deflection profile at transfer and at service

Check of un-cracked behaviour (cracking moment)

- Section is un-cracked if no tension / tension < modulus of rupture $f_{cr} = 0.7\sqrt{f_{ck}} = 4.43$ N/mm².
- Cracking moment $M_{cr} = (f_{cr} + \eta P_i/A + \eta P_i e/Z)Z = (4.43 + 5.64 + 8.12)\times12.5\times10^6 = 227$ kN·m > M_T = 171.9 kN·m ✓ → un-cracked ⇒ gross I valid.

7. Cracked Sections – A Brief Note

When $M > M_{cr}$ the section cracks, the effective I reduces and deflections increase rapidly (stiffness of a Type 3 / partially prestressed member). Our syllabus is for **un-cracked beams**, but one should know:

- **Bilinear (M-δ) behaviour** – before cracking: gross I ; after cracking: cracked-transformed I_{cr}.
- **Effective moment of inertia (Branson)** : $I_{eff} = I_{cr} + (I_g - I_{cr})(M_{cr}/M_a)^3 \le I_g$.
- Deflection then : δ = (deflection by I_{eff}) using net moment $M_{net} = M_T - P\,e$.

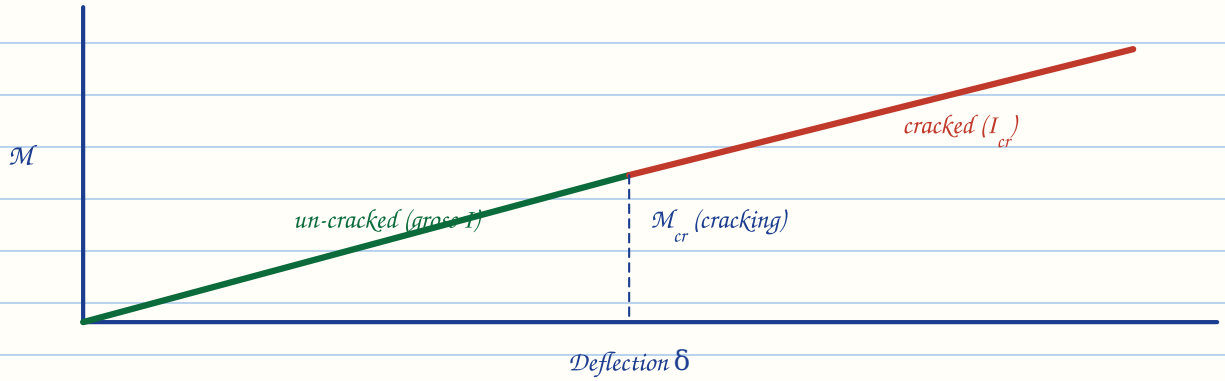


Fig. Moment – deflection curve of a prestressed beam

Transformed section

For accuracy, use the transformed area $A_t = A + (m - 1)\,A_{ps}$ and I_t – the correction is small (≈ 2–5 %). Normal practice is to use the gross section for deflection of un-cracked beams.

Remember : $M_{cr} = Z(f_{cr} + P_e/A) + P_e\,e$ (for the bottom fibre) ; $f_{cr} = 0.7\sqrt{f_{ck}}$ (IS 1343).

Prestressing delays cracking – the member remains stiff (gross I) over the full working-load range ⇒ much **smaller deflections than RCC** of equal span/depth.

8. Prediction of Long-Time Deflections

Under sustained loads concrete creeps and shrinks, steel relaxes and prestress decreases. Deflection and camber change with time. **Long-term deflection ≠ short-term.**

Effects to consider

1. **Creep of concrete** – increases curvature under sustained moment (both load and prestress effects).
2. **Loss of prestress** – reduces the upward camber with time.
3. **Shrinkage** – shortens concrete; with eccentric steel produces additional curvature.
4. **Increase in E_c** with age – reduces deflection slightly (usually neglected).

Method 1 – Effective modulus method (IS 1343 cl. 23.2)

$$E_{ce} = E_c / (1 + \theta)$$

- θ = creep coefficient (IS 1343 : 2.2 – 7 days ; 1.6 – 28 days ; 1.1 – 1 year).
- Long-term deflection = short-term deflection × $(1 + \theta)$, using the same P_e and sustained loads (P_e includes time-dependent losses).
- Net long-term $\delta_{LT} = (\delta_{loads} - \delta_{prestress})\,(1 + \theta)$ with $P = P_e$.

Method 2 – Multiplier method (PCI) for camber/deflection

Stage	Camber (due to P)	Deflection (due to self-weight etc.)
At erection (≈ 30–60 days)	× 1.80	× 1.85
Final (long-time)	× 2.45	× 2.70

- Multipliers applied to the initial (transfer) elastic camber / deflection ; superimposed dead load × 3.0 (final).
- Net final deflection = $2.70\,\delta_g - 2.45\,\delta_p$ + (load not in the initial stage) × $(1 + \theta)$.

Method 3 – Rate of creep / step-by-step (outline)

Divide the time into intervals; compute curvature increments from creep ($\Delta\varepsilon = \theta\,\Delta f/E$), losses and shrinkage – used in computer programs (ACI 209 / CEB-FIP models).

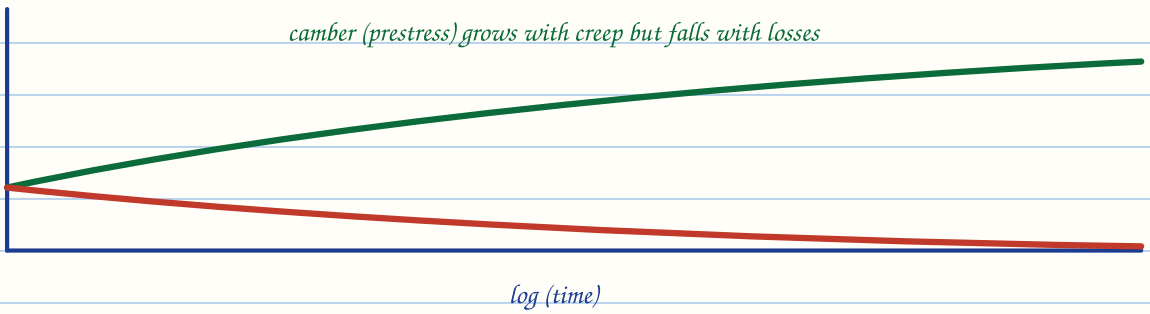


Fig. Time-dependent change of deflection

9. Worked Example – Long-term Deflection

Example 5

For Example 4 (300 × 500, L = 10 m, parabolic tendon, P_i = 1057 kN) compute the long-term deflection after creep and losses. θ = 1.6 (28 days), η = 0.8, all service loads sustained (3.75 + 10 = 13.75 kN/m).

Method 1 – Effective modulus

Short-term at service ($P_e = 0.8P_i$) : δ_w = 15.88 ↓ ; δ_p = 9.38 ↑ ; net = 6.50 mm ↓

Long-term : $\delta_{LT} = (1 + \theta) \times$ net = 2.6 × 6.50 = 16.9 mm ↓

Check : 16.9 mm < $L/250$ = 40 mm ✓ ; $L/350$ = 28.6 mm (after finishes) ✓

Method 2 – PCI multipliers

Initial (transfer) : δ_g = 4.33 ↓ , δ_p = 11.73 ↑ (before losses).

Final = 2.70 × 4.33 – 2.45 × 11.73 + (superimposed 10 kN/m : 11.55 mm) × 3.0 …

= 11.69 – 28.74 + 34.65 = 17.6 mm ↓

Both methods agree (≈ 17 mm ↓) – O.K. Camber at transfer was 7.4 mm ↑ (growing to ≈ 17.1 mm ↑ under self-weight + P at erection × multipliers).

Shrinkage curvature (unsymmetrical steel) : $\kappa_{sh} = \varepsilon_{cs} \cdot m \cdot A_{ps} \cdot e / I_t$; $\delta_{sh} = \kappa_{sh} L^2/8$ (SS beam). For this beam : $\kappa = 3\times10^{-4}\times5.8\times987\times120/3.125\times10^{9} = 6.6\times10^{-8}$/mm → δ = 0.83 mm ↓ (small ✓ ; included roughly in $(1 + \theta)$).

Long-term deflection is typically **2 – 3 times** the short-term deflection ; camber can **reverse** into downward deflection with time if load is heavy.

10. Control of Deflections

Design measures

Measure	How it helps
Increase depth D (span/depth ≤ 20–30)	$I \propto D^3 \rightarrow$ deflection ↓ strongly
Use **high-grade concrete**	↑ E_c, ↓ creep & shrinkage
Adopt **full / limited prestress** (un-cracked)	Gross I valid, camber helps
Adjust **P and e** (profile) properly	Camber balances dead load (load balancing)
Use **parabolic / harped** tendon with high e at mid-span	Larger camber, lower end stresses
Reduce losses : low-relaxation steel, steam curing, good curing	Retains camber/prestress
Provide **precamber** in formwork / compression steel	Offsets long-term deflection
Continuity / composite action / T-section	↑ stiffness, ↓ moment
Use **non-prestressed** compression steel	Reduces creep curvature
Delay application of finishes, partitions; propping	Reduces deflection after fixing finishes
Limit transfer-time camber : control f_{ci}, release	Avoid excessive camber (top cracking)

Span / effective-depth ratios (practical guide)

Member	L/D
Simply supported beams (PSC)	20 – 30
Continuous beams	25 – 35
Slabs	30 – 45
Cantilevers	8 – 12

Deflection of a PSC beam is far less than RCC – but because the members are slender, **always check at transfer and long-term.**

11. Important Questions & Formula Sheet

Short answers

1. State the IS 1343 limits for deflection.
2. List the factors influencing the deflection of a PSC beam.
3. Write the camber for parabolic tendon with end eccentricity.
4. What are the effects of creep and shrinkage on deflection ?
5. What is the effective modulus of elasticity ?
6. Differentiate un-cracked and cracked deflection behaviour.

Long answers

1. Derive the deflection of a simply supported beam with (a) straight eccentric (b) bent (c) parabolic tendon.
2. A PSC beam … calculate the short-term deflection at transfer and at service (numerical).
3. Explain the methods of predicting the long-time deflection of PSC beams. Numerical using $(1 + \theta)$.
4. Describe the measures to control deflection of PSC members.

Formula sheet

Item	Formula
Straight eccentric	$\delta = P e L^2 / 8EI \uparrow$
Bent (mid)	$\delta = P e L^2 / 12EI \uparrow$
Parabolic	$\delta = 5 P e L^2 / 48EI \uparrow$
Parabolic with end e_a	$\delta = P L^2 (e_a + 5e_c) / 48EI$
UDL	$\delta = 5 w L^4 / 384EI \downarrow$
Central point load	$\delta = W L^3 / 48EI \downarrow$
Long-term (IS)	$\delta_{LT} = (1 + \theta)\delta$; $E_{ce} = E_c/(1+\theta)$
PCI multipliers	1.80 / 1.85 (erection) ; 2.45 / 2.70 (final)
IS limits	$L/250$; $L/350$ or 20 mm
E_c	$5700\sqrt{f_{ck}}$ N/mm²

All the best ! Practise at least 2 numericals each for short-term and long-term deflection.